System and Method for Multi-Stage Assessment, Provenance-Aware Scoring, Calibration, and Validation of Artificial-Intelligence Output

Cross-Reference to Related Applications

None.

Federal Sponsorship

No federal sponsorship.

Field

The invention relates to computer-implemented systems for evaluating output generated by artificial-intelligence systems. More particularly, it relates to staged machine assessment pipelines, provenance-aware scoring systems, external scorer versus generator separation, human-machine calibration workflows, and benchmark evidence logging and review.

Background

Existing AI evaluation systems frequently suffer from one or more of the following technical deficiencies:

1. They reduce a generated output to a single scalar score with poor auditability.
2. They fail to preserve whether a score came from deterministic heuristics, self-reporting by the generating model, or a separate scoring model.
3. They conflate generation and evaluation, causing the same model to effectively grade itself.
4. They do not preserve structured disagreement evidence from human raters.
5. They mix different evaluation modes into one ranking table, making benchmark comparisons misleading.
6. They provide little durable infrastructure for storing scorer-agreement cases, false-positive cases, false-negative cases, benchmark runs, and calibration decisions.

There is therefore a need for a computer-implemented system that can assess AI outputs in multiple ordered stages rather than one opaque step, preserve scoring provenance, separate generator and scorer roles, support multiple scoring modes without confusing them, maintain a human reconciliation and evidence record, and produce auditable, exportable results suitable for later validation review.

Summary of the Invention

In one aspect, the invention is a computer-implemented assessment system that receives a prompt, obtains a generated response from an AI model, evaluates the response through a plurality of ordered scoring and integrity stages, computes a composite output, and returns the result together with scoring provenance.

In another aspect, the invention provides selectable scoring modes, including a deterministic heuristic mode, an external secondary scorer mode, and a self-reported inline scoring mode.

In another aspect, the invention records whether scoring was produced by heuristics, a separate scorer, or the generator itself, and surfaces that provenance in both internal data structures and user-facing reports.

In another aspect, the invention provides a human-machine calibration subsystem that identifies disputed items, presents panel and machine scores together, allows a reconciler to record a disposition, optionally stores reconciled scores, and preserves durable calibration history.

In another aspect, the invention provides an evidence subsystem that stores and reviews scorer-agreement cases, benchmark-suite runs, false-positive cases, and false-negative cases.

In another aspect, the invention separates benchmark rankings by scoring mode so that a neutral heuristic baseline is not merged with a model-as-judge or external-scorer comparison.

Brief Description of the Drawings

Figure 1 illustrates an overall system architecture including a client or user interface, wrap API, generator connector, scorer connector, multi-stage pipeline, validation and evidence stores, and reporting or export surfaces.

Figure 2 illustrates an eight-stage pipeline including input analysis, RPM gate, dial adjustment, base LLM call, operator scoring, domain coherence, inversion scan, and output gate or composite generation.

Figure 3 illustrates a scoring-provenance decision flow including heuristic fallback, secondary LLM scoring, self-reported scoring, scorer-provider resolution, and fallback behavior when an external scorer fails.

Figure 4 illustrates a calibration-session workflow including a disputed item queue, human panel scores, machine scores, reconciliation disposition, optional reconciled score snapshot, and session-history persistence.

Figure 5 illustrates an evidence subsystem including scorer-agreement entries, benchmark-suite entries, error-case entries, trend summaries, and filtered review or export.

Figure 6 illustrates separated benchmark tables including a neutral heuristic ranking table and an external-scorer ranking table, while prohibiting mixing scoring modes in a single stakeholder-facing ranking.

Detailed Description

1. System Context

The invention may be implemented as a web service, an internal service, a local service, or another networked or local computing environment.

The system may include request and response schemas, a wrap route, pipeline orchestration, pipeline stages, secondary scorer support, validation routes, evidence-store helpers, and calibration-session helpers.

2. Exemplary Inputs

The system may accept one or more of the following:

- prompt text;
- generator provider identifier;
- optional generator model identifier;
- dial configuration values;
- scoring mode selection;
- optional scorer provider identifier;
- optional scorer model identifier;
- human validation labels;
- calibration notes and reconciliation dispositions; and
- benchmark suite definitions and prompt collections.

In one embodiment, a request object includes explicit scoring controls through fields such as a scoring mode, a scoring provider, and a scoring model.

3. Multi-Stage Assessment Pipeline

In one embodiment, a pipeline runner receives the prompt and executes a sequence of ordered stages.

3.1 Input Analysis Stage The system analyzes the input prompt and extracts structured information used downstream.

3.2 RPM Gate Stage The system computes prerequisite or integrity-related features before or alongside generation. In one embodiment, this stage produces RPM-related intermediate values and integrity scores.

3.3 Dial Adjustment Stage The system applies configurable dial values or sensitivity controls that affect how the subsequent scoring pipeline weights or interprets the response.

3.4 Base Generation Stage The system sends the prompt to a selected generator provider and receives a generated response.

3.5 Operator Scoring Stage The system scores the generated response according to a selected scoring mode. The scoring stage does not require one fixed evaluator.

3.6 Domain Coherence Stage The system converts operator-level signals into grouped coherence values.

3.7 Inversion Scan Stage The system evaluates risk, inversion, or instability axes and may compute additional resistance-related values.

3.8 Output Stage The system produces sub-scores, a composite score, a tiered output classification, provenance information, and optional interpretation or analysis data.

4. Scoring Provenance and Mode Separation

The system supports multiple scoring modes.

4.1 Heuristic Fallback Mode A deterministic structural scoring path is used without asking a model to score the response.

4.2 Secondary Scorer Mode A separate scorer model is invoked to assess the generated response. The scorer may be a different provider or model than the generator. The system records which scorer was used.

4.3 Self-Reported Mode The generator may be permitted to emit inline self-scores, but this path is treated as distinct from both heuristic and external-scorer operation.

4.4 Provenance Output The system stores and surfaces a requested scoring mode, an actual scoring method used, a scorer provider, and a scorer model. This provenance may be preserved in machine output and surfaced in reporting and export layers.

5. External Scorer Resolution

When a secondary scorer mode is requested, the system may attempt to resolve an external scorer provider that differs from the generator provider when one is available.

If the preferred scorer fails or is unavailable, the system may fall back to a same-provider scorer according to policy or fall back to deterministic heuristics according to policy. The system records what actually happened rather than hiding the scoring path.

6. Human Validation and Calibration

The invention may include a human validation subsystem.

6.1 Human Labels Human raters may score prompt-response items using a defined rubric and submit structured labels.

6.2 Disagreement Identification The system may identify items with multiple human ratings and detect disagreement between human raters, human panel means, and machine scores.

6.3 Calibration Session In one embodiment, the system presents a calibration queue that includes a disputed item identifier, prompt, response, machine profile, panel profile, largest-gap operator, and prior calibration-session count.

A reconciler may then enter a disposition, explanatory notes, and optional reconciled operator or domain scores. The system stores this session durably for later audit.

7. Evidence Logging and Review

The invention may include an evidence subsystem.

7.1 Evidence Categories The evidence subsystem may store scorer-agreement cases, benchmark-suite runs, false-positive cases, and false-negative cases.

7.2 Review and Export The system may support filtered evidence review, provider and model summaries, trend summaries over time, and export to machine-readable formats.

7.3 Benchmark Separation The system keeps scoring modes separate in stakeholder-facing rankings, for example by maintaining one table for heuristic baseline and one table for external-scorer runs.

8. Reporting Layer

The system may further include reporting and export surfaces that attach a research-stage notice, scoring provenance, scorer provider or model information, and cautionary wording where heuristic signals are involved.

9. Exemplary Output Data

An output result in one embodiment includes stage records, a composite assessment, a tier or classification output, operator-level values, domain-level values, RPM or prerequisite values, inversion or stability values, provenance fields, and optional narrative interpretation.

10. Example Technical Workflows

Workflow A: Single Prompt Assessment

1. receive a prompt;
2. select a generator and scoring mode;
3. generate a response;
4. score the response;
5. compute a composite output; and
6. return a provenance-aware result.

Workflow B: External Scorer Assessment

1. receive a prompt;
2. select generator provider A;
3. resolve scorer provider B;
4. generate a response with provider A;
5. score the response with provider B;
6. store scorer provenance; and
7. present a result and export.

Workflow C: Human Calibration

1. collect multiple human labels;
2. identify disagreement cases;

3. queue disputed items;
4. present human and machine deltas;
5. record a reconciliation decision; and
6. preserve calibration history.

Workflow D: Stakeholder Benchmark Reporting

1. run a frozen prompt set in heuristic mode;
2. run the same prompt set in external-scorer mode;
3. store results separately;
4. display separate ranking tables; and
5. reconcile disagreement cases through human review.

11. Alternative Embodiments

The invention is not limited to the exact implementation currently described. Alternative embodiments may include different provider connectors, different scoring sub-models, different evidence backends, different user interfaces or no user interface, different benchmark eligibility thresholds, batch or streaming execution, command-line or internal service operation, and additional or fewer structured sub-scores.

12. Technical Benefits

The invention may provide improved auditability of AI assessment results, clearer separation between generator and scorer roles, reduced confusion from mixed benchmark modes, persistent human-machine calibration records, durable evidence review and export, and improved governance for stakeholder-facing comparisons.

These benefits arise from the technical structure of the system rather than from abstract theory alone.